Experiment 4 **Digital Electronics** **Monsoon 2018**

Programmable 1-bit ALU

We will perform all experiments from this experiment onwards using the **Digital Test Kit**, which provides 12 Input Switches, 12 LED displays, each consisting of a RED LED and a GREEN LED, and a Clock Generator that can be run in three possible modes; (i) Manual – operated manually by a Switch, (ii) Auto Fast – generating a free-running fast square wave and (iii) Auto Slow – generating a free-running slow square wave with a time-period of about 2 seconds. This kit enable the functional testing of simple digital circuits to be carried out conveniently by applying the (binary) inputs from the Input Switches and observing the output (binary) variable on the LED Displays.

We will use TTL Multiplexers and CMOS XOR gates as required for the implementation of an Arithmetic and Logic Unit (ALU) capable of performing 8 Arithmetic/Logic functions on 1-bit operands, as selected by the Select bits $S_2\ S_1\ S_0$, as listed in **Table 3.1**. Note that the first 6 functions are Logic functions Zero, NOT, AND, NAND, OR and NOR, generating a single-bit output, while the last two are the Arithmetic functions HALF ADDER and FULL ADDER, generating a 2-bit output. However, in this experiment, we will generate only a 1-bit output ***Y*** for the Arithmetic functions as well, in order to restrict the size of hardware, and as such, the Carry output will not be generated for the Addition operations.

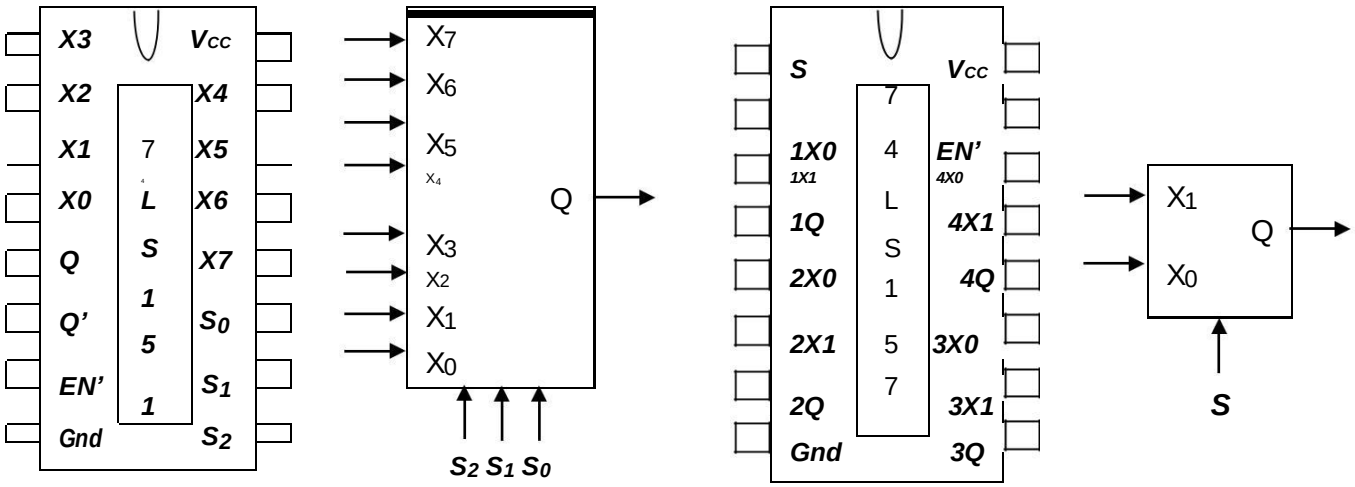
Table 3.1 ALU Function Table

$S_2\ S_1\ S_0$	000	001	010	011	100	101	110	111
ALU Function	**0** (Zero)	A'	$A \bullet B$	$(A \bullet B)'$	$A+B$	$(A+B)'$	A PLUS B	A PLUS B PLUS C

You will use the following ICs (Pin connections given in **Fig. 3.1**) for this experiment:

74LS151 8-input Multiplexer, 74LS157 Quad 2-input Multiplexer, belonging to the TTL family, and CD4070 CMOS Quad 2-input XOR gate (refer to Experiment 1 for its pin connections).

The pin connections and logical symbols of these Multiplexers are given in **Fig. 3.1** below. ***X0***, ***X1***,.....denote the data inputs and ***Q*** denotes the data output for each multiplexer. For a 2-input multiplexer, ***S*** is the Select input common to all the four 2-input multiplexers in the same chip, and $S_2\ S_1\ S_0$ are the select inputs for the single 8-input multiplexer in the chip. Thus for a 2-input multiplexer, the output $Q = X0$ if $S = 0$ and $Q = X1$ if $S = 1$, while for an 8-input multiplexer, the output $Q = Xn$ (selected data input) if $S_2\ S_1\ S_0 = n$ (in binary code). ***EN'*** is the (negative-logic) output enable input, i.e. the corresponding multiplexer output is equal to the selected data input only if ***EN'*** = **0**. ***Q***-outputs of multiplexers are LOW if ***EN'*** = ***1***.



(a) 8-input MUX **(b) 2-input MUX**

Fig. 3.1 Pin Connections and Logical Symbols of Multiplexers

We will use an 8-input multiplexer with the following Data inputs to generate the output ***Y*** as required by the design specifications given in **Table 3.1.**

X0 = ***0***, ***X1*** = ***A'***, ***X2*** = ***A•B***, ***X3*** = ***(A•B)'***, ***X4*** = ***A+B***, ***X5*** = ***(A+B)'***, ***X6*** = ***A⊕B***, ***X7*** = ***A⊕B⊕C***.

A⊕B and ***A⊕B⊕C*** will be generated with two 2-input XOR gates, and ***A'*** will be generated by a third XOR gate. We will use 2-input multiplexers to generate the other four terms, as shown in **Fig. 3.2**. This implementation requires only one chip, whereas any implementation using gates would need more, as a single gate chip contains only one kind of gate (AND/OR/NAND/NOR).

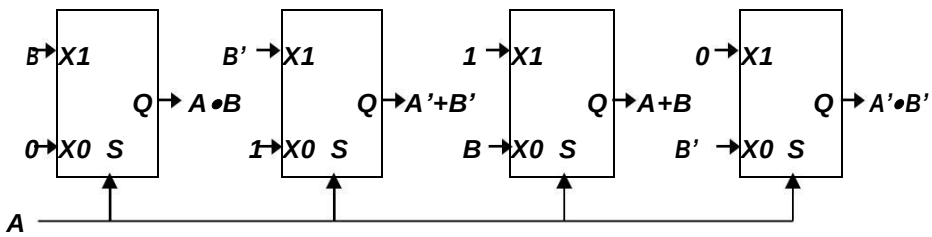


Fig. 3.2 Generation of different functions of A and B using 2-input Multiplexers

1. Recognise the pins provided on the kit for applying inputs to the LED display. Apply ***V_{DC}*** and ***Gnd*** one by one to each of these pins and verify that the RED LED glows for HIGH input, the GREEN LED glows for LOW input, and both LEDs are OFF when no input is connected. Connect the outputs of the Input Switches one by one to the input of an LED display and note the positions of the Input Switches for HIGH and LOW outputs.

2. Apply ***A*** and ***B*** inputs from two **Input Switches** to one of the gates in the XOR chip to generate ***A⊕B***. Generate ***A⊕B⊕C*** by applying ***A⊕B*** and the third Input ***C*** from another **Input Switch** to a second XOR gate. Connect the outputs of the two XOR gates to two LED display inputs. Draw the circuit diagram of these two XOR gates indicating all the pin numbers. Verify the logic of these two outputs for all combinations of values of ***A***, ***B*** and ***C***, as you have already done in **Experiment 2**.

3. Apply the ***B*** input to another XOR gate from the same chip, with the other input of the same XOR gate connected to ***V_{DC}***. Verify that the output of this XOR gate gives ***B'***. Draw the complete circuit diagram of the circuit given in **Fig. 3.2**, including the XOR gate generating ***B'*** from the ***B*** input. Mark all the pin numbers on the circuit diagram. Assemble the complete circuit using the four two-input multiplexers in the 74LS157 chip and two XOR gates in the CD4070 chip by proceeding as follows:
   (i) Connect the ***V_{CC}*** and ***Gnd*** pins of the chip to the ***V_{DC}*** and ***Gnd*** buses on the breadboard.
   (ii) Connect the ***EN'*** input to ***Gnd*** and the inputs marked with ***0*** and ***1*** in ***Fig. 3.2*** to ***Gnd*** and ***V_{DC}*** respectively. Apply the inputs marked with ***B*** and ***B'*** from the XOR gate used in step 2.
   (iii) Apply the **S** (Select) input from the ***A*** input used in step 2 and verify experimentally, by applying all combinations of values of ***A*** and ***B***, that the four outputs do generate the different logic functions AND, NAND, OR and NOR of ***A*** and ***B*** as indicated in **Fig. 3.2**.

4. Test the 8-input multiplexer (74LS151) by proceeding as follows:
   (i) Connect the ***V_{CC}***, ***Gnd*** and ***EN'*** pins of the chip to ***V_{DC}***, ***Gnd*** and ***Gnd*** respectively.
   (ii) Apply the 8 possible combinations of values of the 3-bit Select input ***S_2 S_1 S_0*** from three Input Switches in the binary number sequence and verify that for each value of ***S_2 S_1 S_0*** that the output ***Q*** is given by the input ***Xn*** where ***S_2 S_1 S_0*** = ***n*** (in binary code).
   (iii) Make ***EN'*** = ***1*** and verify that all the multiplexer outputs are ***0***, irrespective of ***A*** and ***B***.

5. Assemble the complete circuit for generating ***Y*** by making connections to the 8 Data inputs ***X0***, ***X1***, ***X2***, ... ***X7*** of the 74LS151 chip, as required by the ALU Function Table given in **Table 3.1**, from the outputs generated by the circuit assembled in step 3. Draw the circuit diagram with pin numbers. Apply the Function Select inputs ***S_2 S_1 S_0*** and the Data Inputs ***A B C*** from six **Input Switches**. Observe and tabulate the output ***Y*** for all combinations of ***S_2 S_1 S_0*** and 3 sets of values of the data inputs ***A B C***. Verify that the results conform to the ALU functions given in **Table 3.1**.